INCH-POUND
MIL-PRF-94/7B
16 February 2001
SUPERSEDING
MIL-PRF-92/7A
22 May 1998

PERFORMANCE SPECIFICATION

RESISTOR, VARIABLE, NONWIRE WOUND PRINTED CIRCUIT BROAD MOUNTING, STYLE RV8

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for style RV8 composition, variable resistors.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-94 - Resistor, Variable, Composition, General Specification for.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-202 - Test Methods Standard Electronics and Electrical Component Parts.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Document Automation and Production Service, Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAT, Post Office Box 3990, Columbus, OH 43216-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification sheets, or MS sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-94.

3.2 Interface and physical dimensions. The resistors shall meet the interface and physical dimensions specified in figure 1.

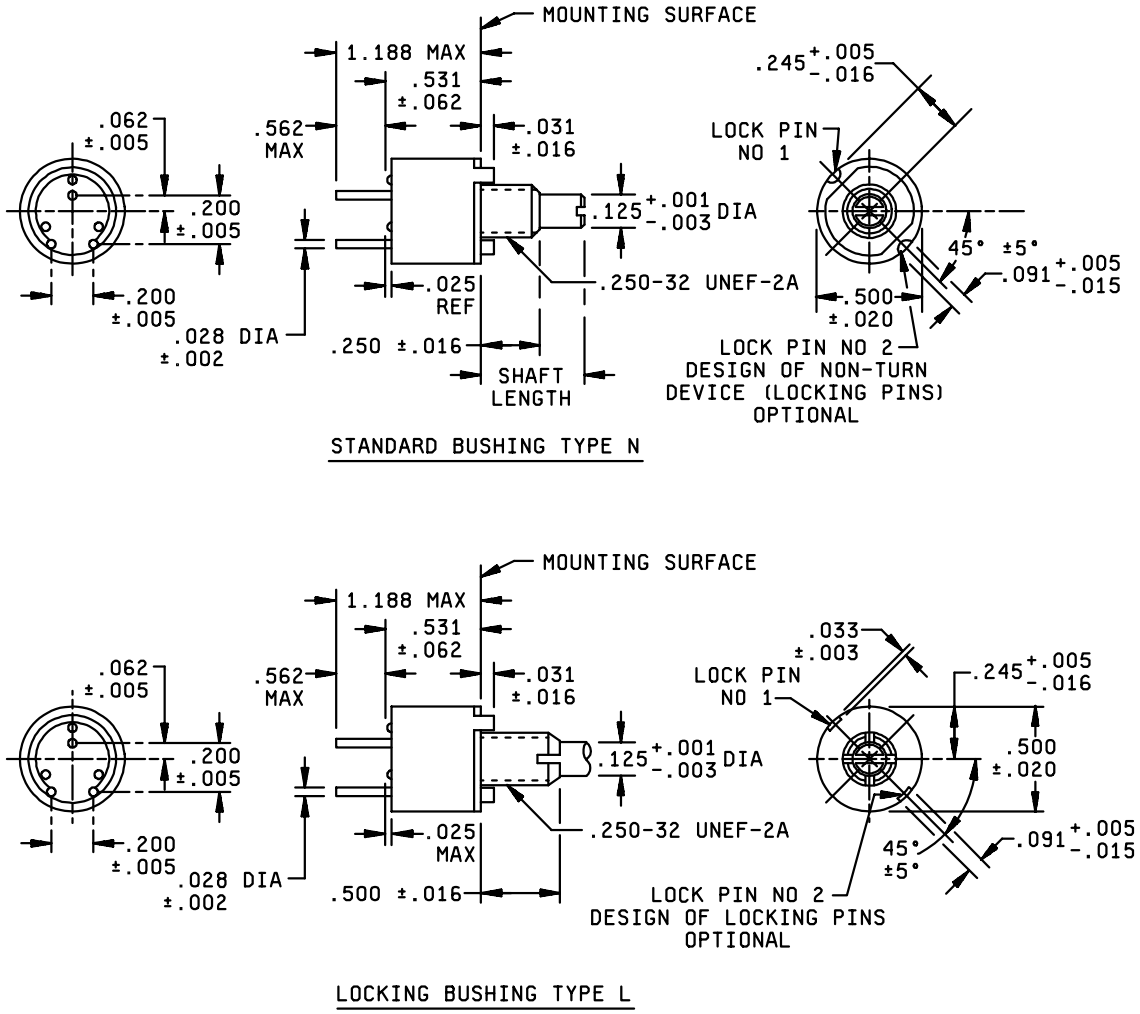
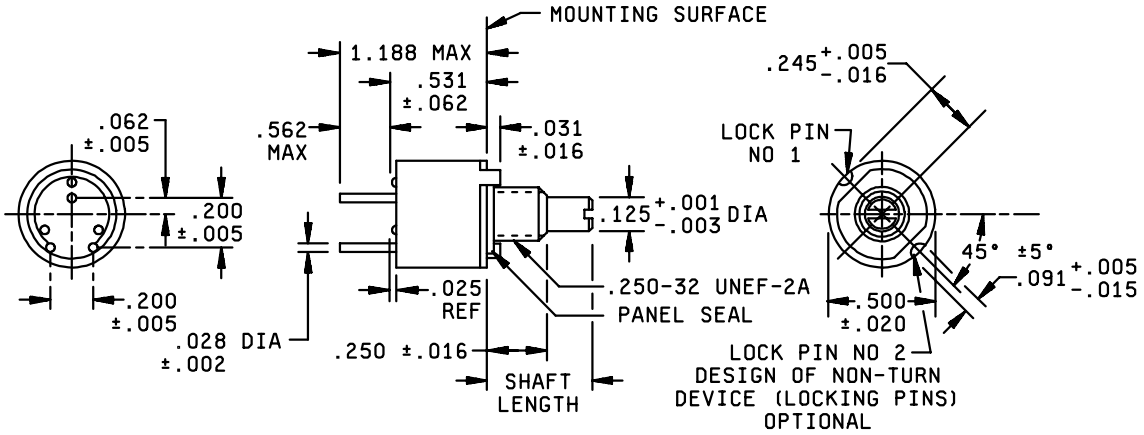
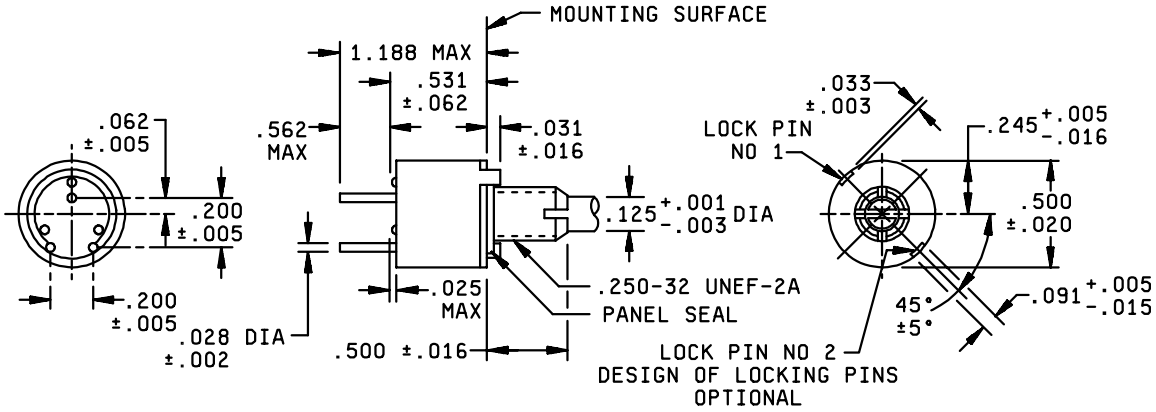


FIGURE 1. Style RV8.

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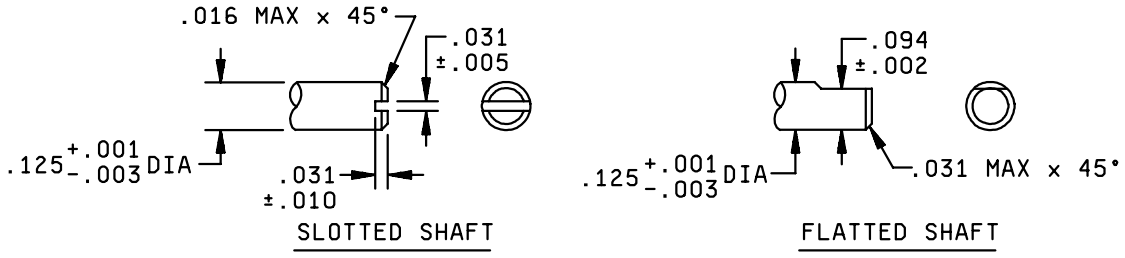
SHAFT AND PANEL SEALED STANDARD BUSHING TYPE S



SHAFT AND PANEL SEALED LOCKING BUSHING TYPE T

FIGURE 1. Style RV8 - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
0.001	0.03	0.025	0.64	0.125	3.18	0.531	13.49
0.002	0.05	0.028	0.71	0.200	5.08	0.562	14.27
0.003	0.08	0.031	0.79	0.234	5.94	0.581	14.76
0.005	0.13	0.033	0.84	0.245	6.22	0.688	17.48
0.010	0.25	0.062	1.57	0.250	6.35	0.812	20.62
0.015	0.38	0.091	2.31	0.453	11.51	1.880	47.80
0.020	0.51	0.094	2.39	0.500	12.70		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information.
3. Unless otherwise specified, tolerance is ±0.016 (0.41mm).
4. When terminals are located symmetrically, the contact terminal shall be identified on the unit. The identifying mark shall be at the option of the supplier.

FIGURE 1. Style RV8 - Continued.

3.3 Nominal total resistance and rated continuous working voltages. The nominal total resistance and rated continuous working voltage (RCWV) shall be as specified in table I.

TABLE I. Nominal total resistance and rated continuous working voltage.

Nominal total Resistance (in ohms)	RCWV 1/ (volts)		Nominal total resistance (in megohms)	RCWV 1/ (volts)	
	Taper A	Tapers C and F		Taper A	Tapers C and F
100	7	5	0.10	224	160
200	10	7	0.20	316	200
250	11	8	0.25	350	200
500	16	11	0.50	350	200
1,000	22	16	1.00	350	200
2,000	31	22	2.00	350	200
2,500	35	25	2.50	350	200
5,000	50	36			
10,000	71	50			
20,000 2/	100	70			
25,000	112	80			
50,000	158	112			

1/ Rated for continuous working voltage at 70°C.
2/ For replacement purposes only. Not for new design.

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3.4 Minimum resistance. The maximum value of minimum resistance shall be 15 ohms for total resistance values of 100 ohms to 500 ohms inclusive.

3.5 Shaft length. The length of the operating shaft shall be in accordance with table II.

TABLE II Shaft length.

Symbol	Length of shafts from mounting surface of resistors, (nominal)		
	Flatted	Slotted	
	Bushings N and S (inches ±0.313)	Bushings N and S (inches ±0.313)	Bushings L and T (inches ±0.313)
A		0.625	0.625
B		0.500	
D	0.875	0.875	0.875
L		0.375	

1/ For resistors with panel seals, the mounting surface shall be considered the metal face of the panel seal when the seal is firmly seated against the resistor body.

3.6 Mounting and locking nuts. The mounting nut shall be 0.078 inches (1.981 mm) thick and measure 0.312 inch (7.925 mm) across the hexagonal flats. The locking nut shall be 0.156 inches (3.962 mm) thick and shall measure 0.312 inches (7.925 mm) across the hexagonal flats. The thread size shall be 0.25-32 NEF-2B.

3.6.1 Internal tooth lockwasher. Internal tooth lockwasher shall be supplied and when mounted, shall have a maximum thickness of approximately 0.045 inch (1.143 mm)..

3.6.2 Retainer rings. If retainer rings are used, shall have a maximum thickness of 0.032 inches (0.813 mm).

3.7 Power rating. The power rating shall be 0.5 watts for taper A resistors and 0.25 watts for taper C and F. Derate 50 percent for nonmetallic panel mounting.

3.8 Torque.

3.8.1 Operating torque. The torque required to effect rotation shall be 0.50 ounce-inch minimum and 6 ounce-inch maximum.

3.8.2 Stop torque. The torque applied to the operating shaft to the stops shall be 3 pound-inch.

3.8.3 Locking torque (as applicable). For the locking bushing type resistors, the locking nut shall be tightened with a torque of 8 pound-inches. After the locking nut is tightened, the contact arm shall not move when a torque of 20 ounce-inches is applied to the shaft.

3.9 Total mechanical rotation. The total mechanical rotation without a switch shall be within the limits of 292 degrees and 298 degrees.

3.10 Thermal cycling. The resistance shall not change in excess of 4 percent.

3.11 Immersion. When resistors are tested as specified in 4.6, no more than four bubbles shall be emitted.

3.12 Resistance to solvents. When resistors are tested as specified in 4.7, there shall be no evidence of mechanical damage and marking shall remain legible.

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-94, except immersion and resistance to solvents testing.

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4.2 Additional qualification inspections. Additional qualification inspections are immersion and resistance to solvent. These shall be added prior to solderability test as specified in table VIII, group IA of MIL-PRF-94.

4.3 Additional group A inspections. Additional group A inspections are immersion (see 3.11 and 4.6); these shall be included in table X, subgroup 1 of MIL-PRF-94 prior to total resistance test.

4.4 Dielectric withstanding voltage. In the dielectric withstanding voltage test, the applied potential shall be 750 volts rms at atmospheric pressure, and 350 volts rms at a reduced barometric pressure.

4.5 Immersion. Resistor shall be immersed for 1 minute ±5 seconds in water at 85°C +5°C, -0°C.

4.6 Resistance to solvents. Resistance shall be tested in accordance with method 215 of MIL-STD-202

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-94 is applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification, and the complete PIN (see 1.2).

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.1).

c. Packaging requirements (see 5.1).

6.3 Extension of qualifications. Qualification to style RV6 will qualify style RV8 with the following tests:

Immersion
Resistance to solvents

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11

Review activities
Army - AR, AT, AV, CR4, MI
Navy - AS, CG, MC, OS
Air Force – 19, 99

Preparing activity:
DLA - CC

(Project 5905-1594-06)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-94/7B	**2. DOCUMENT DATE (YYMMDD)** **01/02/16**

3. DOCUMENT TITLE RESISTORS, VARIABLE, COMPOSITION, STYLE RV8

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Incl Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Supply Center, Columbus ATTN: DSCC/VAM	b. TELEPHONE (Include Area Code) (1) Commercial (614) 692-8754 (2) AUTOVON 850-8754
c. ADDRESS (Include Zip Code) 3990 East Broad Street Columbus, OH 43213-1199	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5803 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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